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(Reference Member State)

DECENTRALISED PROCEDURE

PUBLICLY AVAILABLE ASSESSMENT REPORT FOR A VETERINARY MEDICINAL PRODUCT

BioBos Mastiguard suspension for injection
(CZ, EE, EL, HU, LT, LV, PL, RO, SK)

PRODUCT SUMMARY

EU Procedure number	CZ/V/0189/001/DC
Name, strength and pharmaceutical form	BioBos Mastiguard suspension for injection
Applicant	Bioveta, a.s. Komenského 212/12 683 23 Ivanovice na Hané Czech Republic
Active substance(s)	*Escherichia coli*, strain 5117, J5, inactivated *Staphylococcus aureus,* strain DSM 4910, inactivated
ATC Vetcode	QI02AB17
Target species	Cattle (cows and heifers)
Indication for use	For active immunisation of healthy cows and heifers, in herds of dairy cattle with repeated occurrence of mastitis, to reduce the occurrence of subclinical mastitis, the incidence and severity of clinical symptoms of mastitis caused by mucoid strains of *Staphylococcus aureus* and *Escherichia. coli*.

The Summary of Product Characteristics (SPC), the labelling and package leaflet for this veterinary medicinal product (VMP) is available in the Union Product Database (UPD).

PUBLIC ASSESSMENT REPORT

Legal basis of original application	Full application in accordance with Article 8 (full dossier) of Regulation (EU) 2019/6.
Date of completion of the original decentralised procedure	07/08/2024
Date product first authorised in the Reference Member State (MRP only)	NA
Concerned Member States for original procedure	RMS: CZ CMS: EE, EL, HU, LT, LV, PL, RO, SK

I. SCIENTIFIC OVERVIEW

The product is manufactured and controlled using validated methods and tests, which ensure the consistency of the product released on the market.
It has been shown that the product can be safely used in the target species; the slight reactions observed are indicated in the SPC.
The product is safe for the user, the consumer of foodstuffs from treated animals and for the environment, when used as recommended. Suitable warnings and precautions are indicated in the SPC.
The efficacy of the product was demonstrated according to the claims made in the SPC.
The overall risk/benefit analysis is in favour of granting a marketing authorisation.

II. QUALITY ASPECTS

A. Qualitative and quantitative particulars

Each dose (2 ml) contains:
Active substances:

Escherichia coli, strain 5117, J5, inactivated	RP ≥ 1*
Staphylococcus aureus, strain DSM 4910, inactivated	RP ≥ 1*

*) Relative potency (RP) is determined by comparing the antibody level with the antibody level in serum of mice prepared with a reference batch of vaccine compliant with the challenge test in target animals.

Adjuvant:

Aluminum hydroxide hydrated for adsorption 2%	0.4 ml

Excipients:

Thiomersal	0.2 mg
Formaldehyde	≤ 1 mg
Sodium chloride	
Water for injections	

The container/closure system
Type I glass vials of 10 ml with chlorobutyl elastomer closure and aluminium or flip off caps.
Type II glass vials of 50 or 100 ml with chlorobutyl elastomer closure and aluminium or flip off caps.
Translucent plastic (HDPE) vials of 15, 60 or 120 ml with chlorobutyl elastomer closure and aluminium or flip off caps.

Package size
Plastic box of 10 glass or plastic vials of 5 doses (10 ml)
Cardboard box of 1 glass or plastic vial of 5 doses (10 ml), 25 doses (50 ml), 50 doses (100 ml)

The choice of the active substances, cultivation media, formulation, inactivating agent and adjuvant are justified.

The inactivation process and the detection limit of the control of inactivation are correctly validated.

The product is an established pharmaceutical form and its development is adequately described in accordance with the relevant European guidelines.

B. Method of Preparation of the Product

The product is manufactured fully in accordance with the principles of good manufacturing practice at a licensed manufacturing site.

Process validation data on the product have been presented in accordance with the relevant European guidelines.

C. Control of Starting Materials

Starting materials of non-biological origin used in production comply with indicate pharmacopoeia monographs or in-house specifications.

Biological starting materials used are in compliance with the relevant Ph. Eur. Monographs and guidelines and are appropriately screened for the absence of extraneous agents according to the Ph. Eur.

The master and working seeds have been produced according to the Seed Lot System as described in the relevant guideline and satisfactorily tested according to current European requirements.

D. Control tests during production

The tests performed during production are described and the results of 3 consecutive runs, conforming to the specifications, are provided.

E. Control Tests on the Finished Product

The tests performed on the final product conform to the relevant requirements; any deviation from these requirements is justified. The tests include in particular:

Appearance
Sterility
Airtightness

Extractable volume
pH
Potency*
Potency test of *E. coli* component
Potency test of *S. aureus* component
Identity*
Aluminium Content*
Thiomersal content*
Endotoxins assay **
Formaldehyde content*
*performed as an in-process control on bulk vaccine.
**performed on bacterin just before vaccine bulk formulation

The demonstration of the batch to batch consistency is based on the results of 3 batches produced according to the method described in the dossier. Other supportive data provided confirm the consistency of the production process.

F. Stability

Stability data on the active substances have been provided in accordance with applicable European guidelines, demonstrating the stability of the active substance when stored under the approved conditions.

The storing of the live antigens (*E. coli* and *S. aureus*) during the manufacturing process can be storage at the temperature 2-8 °C up to 72 hours. The inactivated and concentrated antigens (*E. coli* and *S. aureus*) can be storage at the temperature 2-8 °C up to 6 months.

The stability results of the tested three batches of bulk vaccine confirmed storage for up to 1 month at 2 - 8 °C after filling into 10L bottles and before filling into the final primary container.

Stability data on the finished product have been provided in accordance with applicable European guidelines, demonstrating the stability of the product throughout its shelf life (2 years) when stored under the approved conditions at (2 °C – 8 °C).

The in-use shelf-life of the broached vaccine (10 hours) is supported by the data provided.

G. Other Information

Not applicable.

III. SAFETY ASSESSMENT

The vaccine is administered intramuscularly at dose 2 ml to pregnant cows and heifers.
- First injection is 45 days before expected delivery date.
- Second injection is 3 weeks after the first application.
This complete immunization program must be repeated with each pregnancy.

Safety studies have been performed with a vaccine batch with maximum antigen content produced according to the described production process.
Field studies have been performed with a representative vaccine batch produced according to the described production process.

Laboratory trials

The safety of the administration of one dose, the repeated administration of one dose and safety related to reproductive performance was performed as one controlled laboratory study on the basis of Ph. Eur. 5.2.6. which in total included 16 animals (8 vaccinated animals and 8 control animals with placebo administration).

Rectal temperatures, general health status and local reactions were observed.
The safety studies demonstrate that the administration of one dose and the repeated administration of a dose can be considered to be safe, when used in accordance with the recommended vaccination schedule. The average temperature did not exceed a difference of 1.5 °C for all heifers and none of the heifer raised the body temperature by more than 2 °C. No systemic reactions were observed. Local reactions at the vaccine site were obtained for one individual, there was a swelling with a diameter of 0-2 cm after second dose application. The swelling then grew up to average about 2-5 cm but then decreased again. Before the third dose, the swelling gradually decreased.

The observed reactions are reflected in the relevant SPC and package leaflet sections:

Very rare (<1 animal / 10,000 animals treated, including isolated reports):	Injection site swelling[1] Injection site pain[2] Elevated temperature[3]

[1] swelling (on average 5 cm²), which disappears within 1 to 2 weeks at the latest
[2] which will disappear spontaneously within a maximum of 4 days
[3] a slight and transient increase in body temperature of about 1°C, up to 2°C in some dairy cows, may occur in the first 24 hours after the injection

Effects on reproductive performance were examined. No effect on the reproductive performance in vaccinated pregnant heifers and on health status of new born calves was observed thus the following is stated in the SPC and package leaflet:
"Can be used during pregnancy and lactation."

There are no data suggesting that this product might adversely affect the immune system of the vaccinated animal or its progeny therefore a specific study was not carried out.

The vaccine is inactivated and thus the specific tests to be performed for live vaccines are not applicable.

The adjuvant and excipients used are aluminum hydroxide, formaldehyde and thiomersal. The excipient and adjuvants are included in the Appendix of the Commission Regulation (EU) No 37/2010 – the substances that are not subject to determination of residues. For this reason, the presence of the residues was not tested. Based on this information, no withdrawal period is proposed.

No specific assessment of the interaction of this product with other medicinal product was made. Therefore, an appropriate warning in the SPC is included.

Field studies

Two farms were selected for the clinical trial that do not vaccinate animals against any of the mastitis agents for which the vaccine is intended.
A total number of 32 pregnant heifers were selected on two farms for clinical evaluation of the safety and efficacy of the vaccine, with 16 animals available in each of these farms, which were divided into 2 groups (8 vaccinated and 8 controls).

Body temperature measurements (rectal) were performed. The temperature was measured daily for 3 days before the application of the test vaccine, 4 hours after the application and then daily for 4 days in both groups of animals in both breeding facilities. The vaccine met the criteria for safety testing in the clinical trial. The average body temperature in all heifers did not exceed a difference of 1.5 °C and none of the heifers had an increase in body temperature of 2 °C or more (from the average pre vaccination temperature). The standard physiological temperature for cattle is in the range 37,5-39,5 °C.
The evaluation of local and systemic reactions was monitored only in vaccinated groups of animals for 14 days after each injection. No animals enrolled in the clinical trial experienced local injection site reactions following vaccine administration during the study period. No systemic reactions following vaccine administration were observed in any animal enrolled in the clinical trial during the study period.
The last indicator monitored in the safety evaluation of the vaccine was the effect of vaccination on the course of pregnancy and parturition, but also on the viability of newborn calves. No animals were aborted or affected during pregnancy and birth as a result of vaccination. The births of all vaccinated animals in both farms were uneventful, the newborn calves were viable and in good physical condition with no signs of defects or deficiencies that could be attributed to the vaccine tested.
The safety of the administration of vaccine was demonstrated in field conditions in pregnant cows/heifers, confirming results of laboratory studies.

Environmental Risk Assessment

The applicant provided the first phase environmental risk assessment in compliance with the relevant guideline which showed that no further assessment is required. The assessment concluded that there is a negligible risk to the environment associated with use of the vaccine.

Warnings and precautions as listed in the product literature are adequate to ensure safety to the environment when the product is used as directed.

IV. CLINICAL ASSESSMENT (EFFICACY)

The vaccine is intended for active immunisation of herds of healthy cows and heifers, in herds of dairy cattle with repeated occurrence of mastitis, to reduce the occurrence of subclinical mastitis, the incidence and severity of clinical symptoms of mastitis caused by mucoid strains of *Staphylococcus aureus* and *Escherichia coli*. The vaccine is presented as a liquid suspension (2 ml/dose). It should be administered intramuscularly into gluteal area. This vaccine is intended for pregnant heifers and cows. It is applicated in 2 doses – firs dose is applicated 45 days before expected parturition, second dose is applicated 21 days after first dose.
The challenge stains used for the vaccine efficacy testing were selected as different from the vaccinal strains:
The *E. coli* (MSLB 8204) strain was prepared.
The *S. aureus* CCM 4223 (MSLB 3109) strain was obtained from Veterinary Research Institute in Brno.
The challenge strain belongs to the “CCM 4223” type whereas the vaccinal strain belongs to the “DSM 4910” type. The challenge doses were calibrated before initiation of the efficacy testing to ensure the appropriate severity of the challenges.
For the efficacy testing the batches of the vaccine containing the minimum amount of active substances were used.
The field testing used the vaccine with standard antigens content.

Laboratory Trials

The efficacy of the product has been demonstrated in controlled laboratory studies in accordance with the relevant requirements which show that the vaccine reduces:
the occurrence of subclinical mastitis, the incidence and severity of clinical symptoms of mastitis caused by mucoid strains of *Staphylococcus aureus* and *Escherichia. coli*.

The laboratory efficacy study included the following evaluation:
Onset of immunity:
- with challenge 6 days after parturition

Duration of immunity:
- demonstrated based on challenge 150 days after parturition

Onset of immunity:

The vaccinated group of animals received a vaccine with minimal antigen content, leaving the control group without administration. Six days after parturition, all animals were subjected to a challenge experiment. For bacterial challenge with *E. coli* strain WSLB 8204/1, 16 animals (8 vaccinated + 8 control) were used and for bacterial challenge with the *S. aureus* strain WSLB 3109/1, also 16 animals (8 vaccinated + 8 control) were used. The animals were observed for the next 7 days after the administration.

The rectal temperature was measuring. The blood samples were taken from the vaccinated group of animals on the day of vaccination before the vaccine administration and 10 days before expected parturition. The blood samples were taken as well on the day of challenge before the administration of challenge inoculum to the heifers from both groups of animals.

Comparison of vaccinated and control groups of animals:
a) rectal temperature
- temperature score
- number of days with higher physiological temperature and fever
- AUC (ROC) curve of measured rectal temperatures
b) number of clinical symptoms
c) somatic cell numbers (logarithms of AUC)
d) numbers of unsatisfactory milk samples (SB content ≥ 400.000/mL)
e) serological evaluation
- vaccinated group 45 days before parturition x 6 days after parturition
- non-vaccinated group 45 days before parturition x 6 days after parturition
- animals before vaccination
- animals before the challenge experiment

Rectal temperature

E. coli: Statistical comparison of indicators related to rectal temperature revealed that for all parameters evaluated, statistically significant differences were achieved between the vaccinated and non-vaccinated groups at the level of probability $P < 0.05$.

S. aureus: Compared to the E. coli challenge experiment, no significant febrile statuses were observed. Statistical comparison of indicators related to rectal temperature revealed that for all parameters evaluated, statistically significant differences were achieved between the vaccinated and control groups at a level of probability $P < 0.05$.

The typical signs of acute mastitis such as hardening, swelling, redness and other such as milk changes (flakes, clots or watery milk) were evaluated. Each sign was included in the statistical evaluation as one point.

E. coli: All animals in the vaccinated group suffered from symptoms of acute mastitis during the challenge experiment, with the most common symptoms being swelling and hardening of the quarter. Changes in the consistency of the milk also occurred to a large extent in addition to hardening of the quarters and swelling in the non-vaccinated group.
A statistical comparison of the vaccinated and non-vaccinated groups based on the incidence of acute mastitis symptoms, as well as a statistical comparison of the groups based on rectal temperature indicators, resulted in a statistically significant difference at the P <0.05 probability level.
S. aureus: Only 2 animals in the vaccinated group suffered from symptoms of acute mastitis during the challenge experiment, with only changes in milk consistency. No animal suffered from swelling or hardening of the infected quarters. Changes in milk consistency in both animals persisted for 2 days. In the non-vaccinated group, hardening of the quarters, swelling and also changes in the consistency of the milk occurred to a large extent in the animals.
By statistical comparison of the vaccinated and non-vaccinated groups based on the incidence of symptoms of acute mastitis, a statistical difference was achieved for the *S. aureus* challenge strain between the vaccinated and non-vaccinated groups, where the incidence of symptoms was higher in the non-vaccinated group than in the vaccinated group.

The milk samples for the somatic cell count evaluation were taken twice a day at each milking from each quarter separately into 50 ml plastic bottles in an amount of 25 ml with the addition of a preservative compound and the analysis of bottled samples were carried out at accredited laboratories on the Bentley FCM fluoro-optoelectronic analyser by flow cytometry according to the ČSN EN ISO 13366-2. Compliant milk samples contained a maximum of 400.000 somatic cells per 1 mL of milk.

The numbers of non-compliant milk samples for each strain were compared separately for the vaccinated and non-vaccinated groups, but also for the whole vaccine in total, regardless of the used challenge strain. Milk with a somatic cell count higher than 400.000 / mL was considered non-compliant. It was found that the number of non-compliant samples in vaccinated animals decreased by 28.02 %.

A significant increase in somatic cell count was observed in all animals after challenge with both bacterial strains. The higher incidence of somatic cells count, as well as clinical symptoms, was not localized only in the infected quarters, but in many cases affected the entire mammary gland. Even in the observation of somatic cell count, it was found that infection caused by the *E. coli* strain was more aggressive than *S. aureus* infection.

Statistical testing showed a decrease in the mean of somatic cells count in milk. Due to the very high variance of values and the small number of animals, the difference in the number of somatic cells count between the vaccinated and the non-vaccinated group was not statistically significant for any of the used bacterial strains. It was found that a decrease in the somatic cells count in the milk of vaccinated cows challenged with *S. aureus* resulted in a decrease in the number of non-compliant milk samples that had more than 400.000 somatic cells in 1 mL during the study period. For this indicator, testing revealed a statistically significant difference at the level of probability p <0.05.

Serological evaluation

The obtained sera were used for serological evaluation. The aim was to determine that the test item has the potential to stimulate the immune system to produce antibodies against the contained bacterial strains of *E. coli* and *S. aureus*. Evaluation of the antibody response in blood serum was performed by ELISA.

It was confirmed that the amount of antibodies against *E. coli* was several times higher in 10 days before parturition than in the collection before vaccination. It follows that the vaccine stimulates the immune system to make antibodies.

The statistically significant difference was confirmed in the comparison of the groups at the time of blood sampling 6 days after parturition, i.e. on the day of challenge, when the antibody content in the vaccinated group was statistically higher than in the non-vaccinated group. The statistical difference was confirmed for antibodies formation in the vaccinated animals in the comparison of the sampling 45 days before parturition and 6 days after parturition.

It was confirmed that the amount of antibodies against *S. aureus* was several times higher in 10 days before parturition than in the collection before vaccination. It follows that the vaccine stimulates the immune system to make antibodies.

The statistically significant difference was confirmed in the comparison of the groups at the time of blood sampling 6 days after parturition, i.e. on the day of challenge, when the antibody content in the vaccinated group was statistically higher than in the non-vaccinated group. The statistical difference was confirmed for antibodies formation in the vaccinated animals in the comparison of the sampling 45 days before parturition and 6 days after parturition.

Conclusion

Statistical analysis showed a significant decrease in **clinical signs** of acute mastitis after application of both challenge strains in the vaccinated group, which was also shown in the reduction of rectal temperature rise of fever and the number of days when the physiological temperature of animals is above 39.5 °C.

No statistically significant differences were obtained in the evaluation of **somatic cell count**, although the somatic cell count was 24.44 % higher in the *E. coli* strain for the non-vaccinated group than in the vaccinated group, and the somatic cell count was even higher in the non-vaccinated animals by 48.46% in the *S. aureus* strain. Overall, for both strains together, the number of somatic cells in the vaccinated group decreased by 35.21 %, which was reflected in the number of unsatisfactory milk samples by 28.02 %. Despite the reduction in total somatic cell counts in the vaccinated groups, the failure to achieve a statistically significant difference in somatic cell counts was probably due to the high variance of the measured values and the low number of animals included in the study.

It was confirmed that the amount of **antibodies** against both *E. coli* and *S. aureus* was several times higher in 10 days before parturition than in the collection before vaccination. It follows that the vaccine stimulates the immune system to make antibodies, which was confirmed with the serological evaluation where the significant statistical difference between vaccinated and non-vaccinated animals were obtained in the comparison of the sampling 45 days before parturition and 6 days after parturition, i.e. on the day of challenge.

Duration of immunity:

In the efficacy study duration of immunity, the vaccinated group of animals received a vaccine with minimal antigens content, leaving the control group without administration. Five months (150 days) after parturition, all animals were subjected to a challenge experiment according to their classification with either *Escherichia coli* strain WSLB 8204/1 or *Staphylococcus aureus* strain WSLB 3109/1 and the animals were observed for the next 7 days after the administration.

Statistical analysis showed a significant decrease **in clinical signs** of acute mastitis after application of both challenge strains in the vaccinated group, which was also shown in the reduction of rectal temperature the number of days when the physiological temperature of animals is above 39.5 °C. Statistically significant differences were obtained in the evaluation of **somatic cell count** based on the non-compliant samples (samples with somatic cells content more than 400 000/mL of milk), where the number of non-compliant samples decreased by 54.69 % in vaccinated animals after *E. coli* challenge and by 33.79 % after *S. aureus* challenge. Overall, for both strains together, the number of non-compliant samples in the vaccinated group decreased by 44.24 %.

It was confirmed that the number of **antibodies** against *E. coli* and *S. aureus* was several times higher 6 days after parturition than in the sampling before vaccination. It follows that the vaccine stimulates the immune system to produce antibodies, which was confirmed by serological evaluation, where there is a significant statistical difference, which coincides with the results found in a previous study of the onset of immunity.

Despite the decrease in the level of antibodies, a significantly higher level of antibodies was demonstrated 150 days after parturition than before vaccination of the animals, which indicates the ability of the vaccine to protect animals from mastitis caused by bacterial strains of *E. coli* and *S. aureus* up to 5 months after parturition.

Field Trials

The results from vaccination-challenge trials conducted under laboratory conditions have been supplemented with data from field studies.
The clinical evaluation was conducted at two independent test sites.
The clinical evaluation of the safety and efficacy, a total of 32 animals from 2 farms (A1, A2), with 16 animals available in each of these farms, which were divided into 2 groups (8 vaccinated and 8 controls).

The efficacy of the tested vaccine was evaluated on the basis of the amount of somatic cells and myeloperoxidase enzyme activity in milk samples, but also on the basis of the antibody response of the immune system of vaccinated animals against individual antigens contained in the vaccine.
During the course of the clinical trial, exceptional milk samples were taken outside of the established sampling. These were taken mainly from those mammary gland quarters where signs of acute mastitis were detected. The analysis of the abnormal samples was intended to identify the causative agent of mastitis and thus contribute to the evaluation of the efficacy of the investigational vaccine.

During the clinical trial, a total of 1220 milk samples were collected for somatic cell evaluation and 136 emergency samples for detection of the pathogen causing mastitis. Samples confirming *E. coli* (31 confirmed samples out of 1220 samples collected) accounted for 2.54% and *S. aureus* (15 confirmed samples out of 1220 samples collected) accounted for 6.97% of the total. The bacterial strain *S. uberis* (19 confirmed samples out of 1220 samples taken) was represented by 1.56 %. Thus, mastitis samples accounted for 11.23 % of the total number of samples collected. The remaining 88.77 % of the samples did not have significantly elevated somatic cell counts due to mastitis.
The results show that while the dominant mastitis strain in the A1 farm was caused by *S. aureus*, the dominant mastitis strain in the A2 farm was caused by *E. coli*. The results further indicate that the appropriate animal breeds were chosen for the clinical evaluation.
The increase in somatic cell count was observed in both farms mainly in control animals. The high somatic cell counts in the control animals persisted for a long period of time (up to 7 consecutive sampling points), while in the control animals higher counts were detected at a maximum of 1 sampling point. Somatic cell counts in the vaccinated group reached maximum values of lower millions of somatic cells, whereas in the control group they were higher millions to tens of millions of somatic cells. There was also sampling of non-milk-like secretions that could not be analyzed and therefore these samples were labelled as NA. However, these were only 2 samples obtained from control animal from breed A1. Mastitis occurred very sporadically in vaccinated animals in breed A1, in breed A2 no vaccinated animal suffered from mastitis during the observation period.
During the observation period, there was a statistical difference in the number of somatic cells between the vaccinated and the control group in both farms A1 and A2. This difference was due to the higher frequency, severity and duration of mastitis in the control animals compared to the vaccinated animals.

Statistical processing of the results of myeloperoxidase activity confirmed that the activity of the enzyme increases in mastitis. In the evaluation of the samples, it was also found that there were statistically significant differences in myeloperoxidase activity between the vaccinated and control groups in both breeds. Thus, myeloperoxidase enzyme activity can be considered as an indicator of ongoing mastitis disease, while the efficacy of the vaccine was demonstrated, with vaccinated animals having lower levels of enzyme activity than control animals, whose increased activity was due to the occurrence of mastitis during the clinical trial, while vaccinated animals had almost no mastitis.

Prior to the start of the clinical trial, antibody levels against individual strains were at low levels. A large increase in antibodies was observed between revaccination and the onset of immunity, i.e., 6 days after delivery, when there was a sharp increase in levels against *E. coli* up to 80% and *S. aureus* up to 110%. During follow-up, levels gradually decreased and the downward trend continued until the end of observation.
Statistical analyses were also performed to compare the levels of antibodies against both strains in vaccinated animals before vaccination, 6 days after birth (onset of immunity) and at the last sampling (6+126 days after birth) in both breeds. In all cases of comparison, statistically significant differences in antibody levels for both strains were confirmed, i.e. a statistically significant increase in antibodies at the time of onset of immunity compared to pre-vaccination levels was found, but also a statistically significant decrease in antibodies at the last collection compared to the levels at the time of onset of immunity was found during the clinical evaluation, but despite this, antibody levels in vaccinated animals after the last collection were statistically significantly higher than before vaccination. These results confirm that the vaccine used is effective and stimulates the body to produce antibodies against bacterial strains of *E. coli* and *S. aureus* that have a protective potential for at least 5 months.

The results obtained in this clinical study support the findings obtained in the laboratory trials.

V . OVERALL CONCLUSION AND BENEFIT– RISK ASSESSMENT

The data submitted in the dossier demonstrate that when the product is used in accordance with the Summary of Product Characteristics, the risk benefit profile for the target species is favourable and the quality and safety of the product for humans and the environment is acceptable.

POST-AUTHORISATION ASSESSMENTS

The SPC and package leaflet may be updated to include new information on the quality, safety and efficacy of the veterinary medicinal product. The current SPC is available in the Union Product Database (UPD).